INCH-POUND

MIL-DTL-55302/55G
w/AMENDMENT 1
11 May 2011
SUPERSEDING
MIL-DTL-55302/55G
3 February 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, SOCKET CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

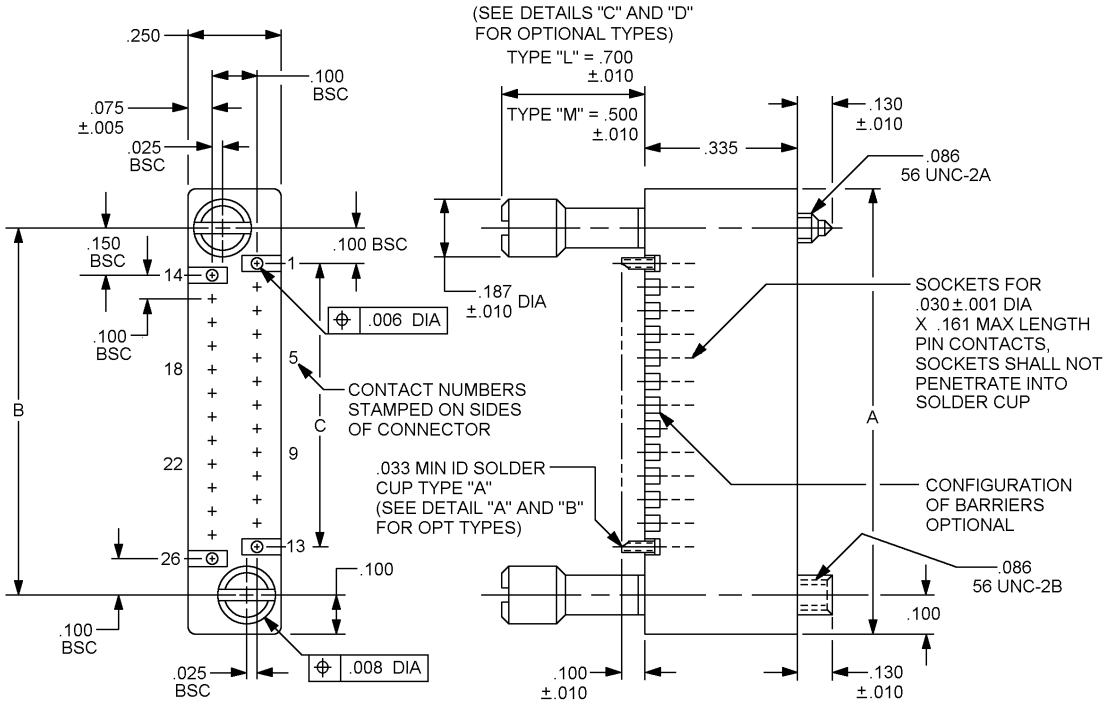
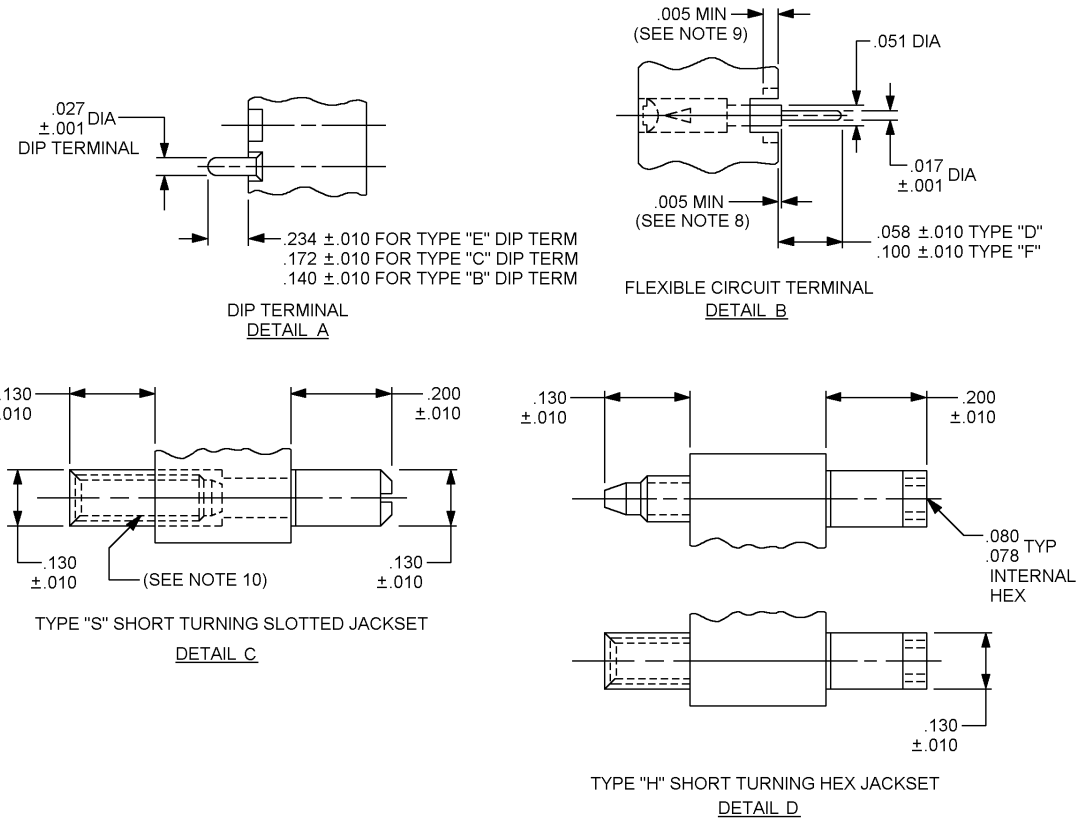


FIGURE 1. Connectors, plug, .100 (2.54 mm) spacing.

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Number of contacts	Dimensions A ±.010	BSC B	REF C	Number of contacts	Dimensions A ±.010	BSC B	REF C
10	.850 (21.59)	.650 (16.51)	.400 (10.16)	44	2.550 (64.77)	2.350 (56.69)	2.100 (53.35)
14	1.050 (26.67)	.850 (21.59)	.600 (15.24)	50	2.850 (72.39)	2.650 (67.31)	2.400 (60.96)
20	1.350 (34.29)	1.150 (29.21)	.900 (22.86)	54	3.050 (77.47)	2.850 (72.39)	2.600 (66.04)
24	1.550 (39.37)	1.350 (34.29)	1.100 (27.94)	56	3.150 (80.01)	2.950 (74.93)	2.700 (68.58)
26	1.650 (41.91	1.450 (36.83)	1.200 (30.48)	60	3.350 (85.09)	3.150 (80.01)	2.900 (73.66)
30	1.850 (46.99)	1.650 (41.91	1.400 (35.56)	66	3.650 (92.71)	3.450 (87.63)	3.200 (81.28)
36	2.150 (54.61)	1.950 (49.53)	1.700 (43.18)	70	3.850 (97.79)	3.650 (92.71)	3.400 (86.36)
40	2.350 (56.69)	2.150 (54.61)	1.900 (48.26)				

FIGURE 1. Connectors, plug, .100 (2.54 mm) spacing - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.051	1.30	.171	4.34
.005	0.13	.058	1.47	.172	4.37
.006	0.15	.075	1.91	.187	4.75
.008	0.20	.078	1.98	.200	5.08
.010	0.25	.086	2.18	.234	5.94
.017	0.43	.100	2.54	.250	6.35
.025	0.64	.108	2.74	.335	8.51
.027	0.69	.130	3.30	.500	12.70
.030	0.76	.140	3.56	.700	17.78
.035	0.89	.150	3.81		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-DTL-55302/56.
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector. As and option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked.
6. Termination layout on .025 (0.64 mm) modular grid.
7. Metric equivalents are in parentheses.
8. Conformal coating clearances shall be .005 (0.13 mm) minimum (not required for optional design).
9. Optional design conformal coating clearance.
10. Hole depth to female guide hardware is .282 min (7.16 mm). Full thread depth to female threaded hardware is .240 min (6.1 mm).

FIGURE 1. Connectors, plug, .100 (2.54 mm) spacing - Continued.

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Requirements:

Dimensions and configuration: See figure 1.

Material: Jackscrews and jack sockets shall be made of corrosion resisting stainless steel in accordance with ASTM A581/A581M or ASTM A582/A582M passivated in accordance with SAE-AMS2700 or shall be corrosion resistant copper nickel alloy (61 ±2% nickel, 2.5% max. other, and the balance zinc and shall have a brinell hardness of 145 to 175.

Plating: The contact plating shall be in accordance with MIL-DTL-55302 or gold in accordance with ASTM B488 type II, code C, class 1.27, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with ASTM B488 type II, code C, class 0.51, or type III, code A, class 0.51 over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 50 to 150 microinches in the termination area.

Contact identification: See figure 1.

Contact engagement and separation forces (number 22 contacts):

Minimum separation: 0.5 ounce.
Maximum engagement: 12.0 ounces for standard force and 4.0 ounces for low force.

Mating and unmating:

Standard insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.56 and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Low insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0.04 times the number of contacts and shall not exceed the maximum insertion force.

NOTE: Not applicable when mating and unmating is accomplished through the engaging of turning jacksets.

Jackscrew and jack socket torque: 2-inch pounds maximum, fully mated.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

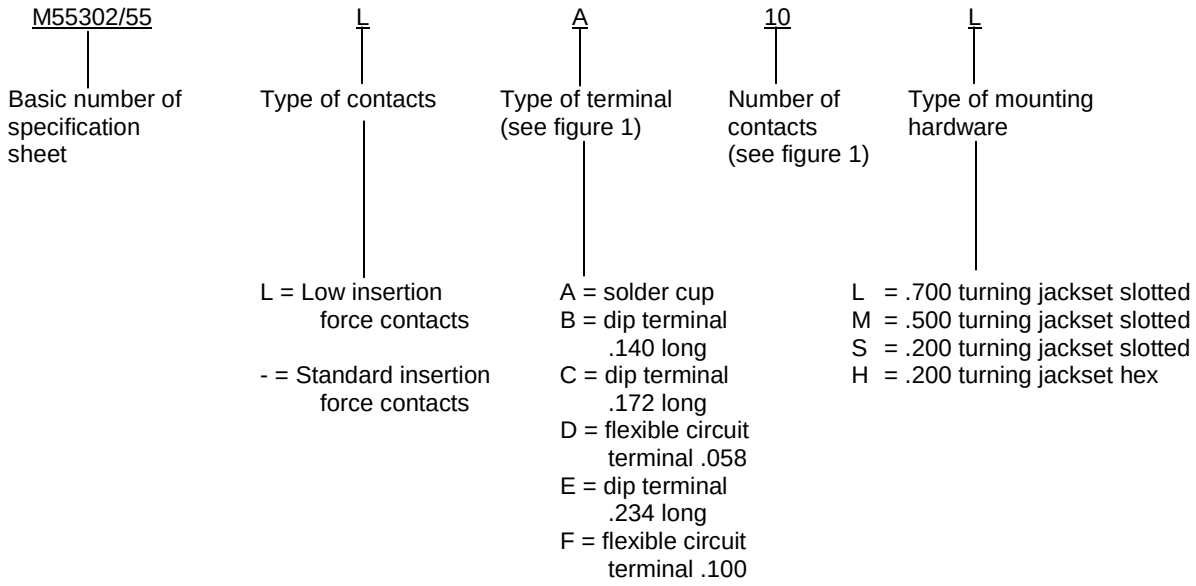
Dielectric withstanding voltage:

Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Current rating: 5.0 amperes maximum in accordance with contact.

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Example of Part or Identifying Number (PIN):



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/56
ASTM A581/A581M
ASTM A582/A582M
ASTM B488
SAE-AMS2700
SAE-AMS-QQ-N-290

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2010-159)

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, MC, OS, TD
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/ .

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